

Experiment 10: Map Coloring Problem

Aim

To implement the **Map Coloring Problem** using the **Constraint Satisfaction Problem (CSP)** approach in Artificial Intelligence.

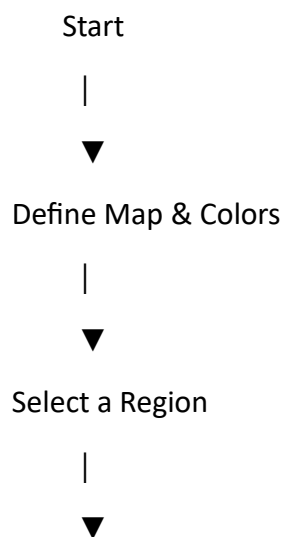
Objective

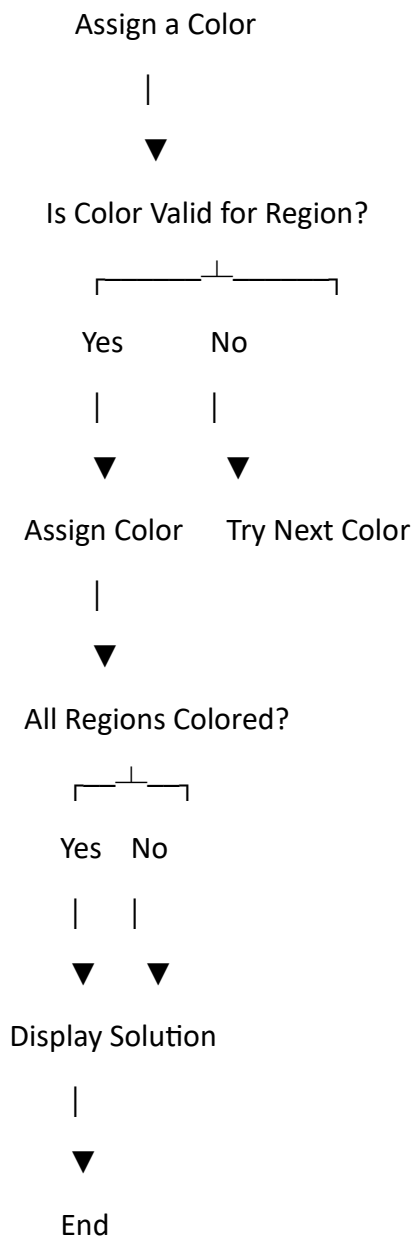
To assign colors to different regions of a map such that no two adjacent regions have the same color while using the minimum number of colors.

Algorithm

1. Define the map as a graph where each region is represented as a node.
2. Specify the adjacency relationship between regions.
3. Define the available colors.
4. Assign a color to the first region.
5. Check whether the assigned color conflicts with adjacent regions.
6. If there is no conflict, assign the color and move to the next region.
7. If a conflict occurs, try another color (Backtracking).
8. Repeat until all regions are colored.
9. Display the color assigned to each region.

Flowchart





Python Code

```
graph = {  
    'A': ['B', 'C', 'D'],  
    'B': ['A', 'C'],  
    'C': ['A', 'B', 'D'],  
    'D': ['A', 'C']  
}
```

```
colors = ['Red', 'Green', 'Blue']
```

```
result = {}
```

```
def is_safe(node, color):
```

```
    for neighbour in graph[node]:
```

```
        if neighbour in result and result[neighbour] == color:
```

```
            return False
```

```
    return True
```

```
def solve(node_list, index):
```

```
    if index == len(node_list):
```

```
        return True
```

```
    node = node_list[index]
```

```
    for color in colors:
```

```
        if is_safe(node, color):
```

```
            result[node] = color
```

```
            if solve(node_list, index + 1):
```

```
                return True
```

```
            del result[node]
```

```
    return False
```

```
nodes = list(graph.keys())
```

```
if solve(nodes, 0):  
    print("Color Assignment:")  
    for node in nodes:  
        print(node, "->", result[node])  
else:  
    print("No solution exists.")
```

Sample Output

Color Assignment:

A -> Red

B -> Green

C -> Blue

D -> Green

Result

Thus, the **Map Coloring Problem** was successfully implemented using the **Constraint Satisfaction Problem (CSP)** approach. The map was colored such that no two adjacent regions had the same color using the minimum number of colors.